

Echo Refocusing: When a Single Pulse Cancels TERA-Space Drift, and When It Provably Cannot

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Status: Working paper, not peer reviewed. Every number in §3–§5 comes from a deterministic, seeded simulation (`packages/libaamt/experiments/echo_refocus.py`, seed 20260708); every number in §7 comes from a real-data measurement over this paper series’ own mark-down sources (`maaiam-alchemist/apps/desktop/scripts/measure_ocm_drift_coherence.py`, seed 20260711); every number in §9 comes from a follow-up investigation of §7’s own finding (`maaiam-alchemist/apps/desktop/scripts/investigate_negative_autocorr.py`, seed 20260711) — all reproducible by running the scripts directly, the same standard EXP-01 set for its lattice sweeps.

Abstract

A prior research pass on “mirror magic” for this architecture ranked five mechanisms; the top four are built and shipped (WP-23 sections 10–12). The fifth — phase-conjugate mirrors, the nonlinear-optics device that reflects a distorted wavefront so its aberrations cancel on the return pass — was graded **research-only**, not because it lacked a computational analog, but because the physical claim underneath it has a precondition that is easy to skip past: a phase-conjugate mirror, like the Hahn spin echo it is mathematically identical to, only refocuses distortion drawn from a *static, retraced* channel (NMR’s inhomogeneous T_2^* dephasing); it does nothing for genuinely random, memoryless noise (homogeneous T_2 dephasing) — textbook NMR, not this paper’s invention, but a distinction worth re-deriving before importing it. This paper tests that precondition directly, in this architecture’s own coordinate space, before building anything on top of it. The operator under test is a **point inversion** about a target, $x \mapsto 2c - x$ — the correct analog of a spin echo’s 180° pulse, and deliberately *not* called a Householder reflection: WP-23 §10.1 already had to correct exactly this confusion once (the antipodal map is orientation-preserving in even dimension, not a true mirror), and conflating the two here would repeat it. Two drift regimes are simulated over the 4D TERA cube — a **coherent** regime (each trial carries one persistent per-trial bias, the computational stand-in for a static field offset) and an **incoherent** regime (zero-mean i.i.d. noise only) — in both unconstrained $\mathbb{R}^4$ and the substrate’s own clamped $[0, 1]^4$. The stated, falsifiable prediction: the pulse should substantially cut final drift in the coherent regime and do essentially nothing in the incoherent one. Measured (50,000 trials per cell): the pulse reduces mean final distance-to-target by **83.6%** (unconstrained) and **80.5%** (TERA-clamped) in the coherent regime ($z > 500$ both cases), and by **0.1%** and **0.5%** respectively in the incoherent regime — a difference indistinguishable from zero at unconstrained scale and a small but real boundary-clamping artifact at TERA scale, characterized in §5 rather than hidden. The precondition holds, precisely where physics says it should and not elsewhere. What this paper does **not** do is claim any

real subsystem in this platform actually produces coherent drift — the most plausible candidate (WP-22’s session visit-history pull) is named in §6 as the next thing to measure, not asserted as already true. §7 runs that measurement — real prose from this paper series’ own markdown sources, projected through WP-22’s actual lexicon TERA projector, tested for lag-1 autocorrelation via a 5,000-draw permutation test — and finds neither of this paper’s two modeled regimes: large, overwhelmingly significant ($p < 10^{-4}$, all seven sessions, every axis) **negative** autocorrelation, a mean-reverting/oscillating structure §3’s model never considered. Real, not a null result, and not the coherent bias §6 hypothesized either — stated as what it is rather than forced into either prior bucket. §9 then investigates *why*, and the answer corrects §7’s framing rather than confirming it: differencing any sequence mechanically manufactures lag-1 autocorrelation of -0.5 regardless of temporal structure, three content- and order-blind controls (pure noise, real paragraphs in random order, word salad) reproduce §7’s exact magnitude under its own test, and a corrected permutation null — shuffling the raw sequence rather than the already-differenced increments — finds the real document order statistically indistinguishable from a random shuffling of the same paragraphs ($p = 0.283$ pooled). The negative number is real; the “real prose alternates register” reading of it is not supported once the null model is fixed.

1. From a lyric to a precondition

The research pass this paper closes out began as a plain-language request: built-in redundancy, instant writing, instant gating, “a bad reflection disperses itself” — and a note that the underlying math should be real, not decorative. Four of the five ranked mechanisms translated directly into shipped code because their governing math has no precondition to check: a Householder reflection is exact regardless of what is reflected (WP-23 §10); a systematic error-correcting code either recovers a codeword or it doesn’t (WP-23 §11); a join-semilattice merge is commutative by construction once the tiebreak is content-complete (WP-23 §12). Phase-conjugate mirrors are different in kind. The physical mechanism — $E^*(r, t)$ propagating backward through the same aberrating medium and undoing what $E(r, t)$ picked up — depends on the medium being the *same* medium on both passes. That is a claim about the *source of distortion*, not about the reflection operator itself, and it was withheld from the “buildable this week” tier for exactly that reason: an operator that is mathematically exact but applied to the wrong kind of input does nothing, or worse, looks like it works on the one example checked and fails everywhere else. This paper checks it properly, the same way EXP-01 checked $\xi_{50} \propto \lambda$ before publishing it and found $\propto \sqrt{\lambda}$ instead.

2. The operator: a point inversion, not a mirror

Define, for a center c and position x in the TERA cube,

$$P_c(x) = 2c - x.$$

Two structural facts, mirroring how WP-23 §10.1 stated the Householder reflection’s:

- P_c is an exact involution, $P_c(P_c(x)) = x$, with the same no-numerical-drift property as H in unconstrained arithmetic.
- P_c preserves distance to c exactly: $\|P_c(x) - c\| = \|x - c\|$. This is the property that matters here — a Hahn echo’s 180° pulse does not distort a spin’s phase magnitude, it negates its sign relative to a reference, which is precisely what P_c does to a displacement vector.
- P_c is orientation-preserving in this (even, 4D) space, $\det = (-1)^4 = +1$ — a rotation by π about c in each coordinate pair, not a chirality-reversing mirror. This is the exact object WP-23 §10.1 named the “antipodal map” and distinguished from the true Householder

reflection $H(v) = (1 - v_T, v_E, v_R, v_A)$ used for the chirality gate. Using P_c here is *correct*, not an oversight: a spin-echo pulse is physically a rotation, not a parity flip, so the math this paper needs is the one WP-23 already flagged as the wrong tool for *redundancy* — and the right tool for *this*.

3. The model

Two independent binary factors, four cells:

Drift regime. - *Coherent*: each of the N simulated trials draws one persistent bias vector $\mu \sim \mathcal{N}(0, \sigma_\mu^2 I_4)$, held fixed for that trial’s entire walk — the computational stand-in for a static field offset (the “same medium” a phase-conjugate mirror or spin echo retraces). - *Incoherent*: no persistent term; every step draws independent noise with no memory of prior steps or of any per-trial identity.

Space. - *Unconstrained*: the walk lives in $\mathbb{R}^4$, isolating the pure mechanism from any boundary interaction. - *TERA-clamped*: position is clamped into $[0, 1]^4$ after every step (the substrate’s own `clampTera4` rule, `packages/libaamt/src/core.cpp`), so the boundary nonlinearity a real TERA-space application would face is present exactly as it would be in production.

Walk. Each trial starts at the cube center $x_0 = (0.5, 0.5, 0.5, 0.5)$, which is also its own target $c = x_0$ — the question asked is “how far from where I started do I end up.” At each of 40 steps, $x \leftarrow x + \mu + \varepsilon_t$ with $\varepsilon_t \sim \mathcal{N}(0, \sigma^2 I_4)$ ($\sigma = 0.01$, $\sigma_\mu = 0.01$ where applicable), clamped if the space condition requires it. The pulse condition applies $x \leftarrow P_{x_0}(x)$ once, at step 20 (the walk’s temporal midpoint — the same placement a Hahn echo’s π -pulse gets, halfway between excitation and readout), with the no-pulse condition otherwise identical. 50,000 trials per cell, seed 20260708.

Prediction, stated before running. In the coherent regime, negating the accumulated bias-driven displacement at the midpoint should cause the *same* bias to re-accumulate through the second half with the opposite net effect, canceling the coherent term exactly in unconstrained space (a short derivation: with $x_{20}^- \approx x_0 + 20\mu$, the pulse gives $x_{20}^+ = 2x_0 - x_{20}^- \approx x_0 - 20\mu$, and 20 further steps of the same μ bring $x_{40} \approx x_0 - 20\mu + 20\mu = x_0$ — the coherent term is gone, leaving only the noise). In the incoherent regime, P_{x_0} preserves distance to the target exactly and future steps are independent of position by construction, so the pulse should change nothing. The boundary clamp can only ever help, never hurt, since it can only pull excursions back toward the cube.

4. Results

space	regime	pulse	$\pm$ se	control	$\pm$ se	reduction	z
unconstrained	coherent	0.12466	0.00020	0.75927	0.00124	83.58%	505.9
unconstrained	incoherent	0.11882	0.00019	0.11898	0.00019	0.13%	0.59
tera-clamped	coherent	0.12403	0.00020	0.63636	0.00071	80.51%	694.6
tera-clamped	incoherent	0.11856	0.00019	0.11919	0.00019	0.53%	2.33

The prediction holds cleanly on the side that matters most: the pulse cuts mean final drift by roughly 5-to-6x in the coherent regime (control/pulse ≈ 6.09 unconstrained, ≈ 5.13 TERA-clamped), in both space conditions, at overwhelming significance. In the incoherent, unconstrained cell — the cleanest test of “does this operator do anything to pure noise” — the effect is 0.13%, statistically

indistinguishable from zero ($z = 0.59$). That is the core result: **the precondition is real, and this operator respects it precisely**, not approximately.

5. The one honest wrinkle: clamping is not quite an isometry

The TERA-clamped incoherent cell shows a 0.53% reduction at $z = 2.33$ ($p \approx 0.02$) — technically “significant” at $N = 50,000$, and worth explaining rather than rounding to zero. P_c preserves distance to c *exactly* only in unconstrained space; once positions are clamped into $[0, 1]^4$ after every step, P_c composed with clamping is no longer a perfect isometry near the boundary — `clamp(2c - x)` can differ from what an unclamped reflection would give whenever the pre-image excursion sits near an edge. At $\sigma = 0.01$ per step over 40 steps, trials rarely approach the boundary (a 4-axis walk starting at the center with this step size stays within roughly ± 0.06 of center for the large majority of trials), so the effect is small — half a percent, not the 80-percent scale of the genuine coherent-regime cancellation — but it is a real, structural asymmetry from the boundary, not sampling noise dressed up as one, and $N = 50,000$ trials is exactly what makes a small true effect visible instead of buried. Read together with unconstrained incoherent’s clean null, this pins the boundary — not the operator — as the source: an application built on this mechanism in the real, clamped substrate should expect a small edge-proximity artifact and can size it, rather than being surprised by it later.

6. What this licenses, and what it does not

This paper validates a **mechanism**: an operator, on this coordinate space, behaves exactly as the phase-conjugate-mirror / spin-echo analogy requires, with the precondition (coherent vs. incoherent source) cleanly separating the cases where it helps from the cases where it is inert. It does **not** establish that any live subsystem in this platform produces coherent drift for it to act on — that is a separate, unmeasured claim, and this paper is deliberately not making it. The most plausible real candidate, named honestly as a hypothesis rather than a result: WP-22’s Odu coverage map tracks a session’s own visit history, and a session that keeps landing near already-visited cells is, by definition, exhibiting a *persistent, non-random pull* toward a region — structurally the “coherent bias” this paper’s simulation models synthetically, not noise. Whether that pull is large enough, and stable enough across a session, to be worth refocusing with a P_c pulse instead of (or alongside) WP-22’s existing escape heuristic is an empirical question this paper has not asked: it would require instrumenting a real session’s TERA-coordinate sequence and testing it for exactly the coherent-vs-incoherent structure §3 defines synthetically here — an autocorrelation or persistent-direction test on real visit data, not assumed. Until that measurement exists, wiring P_c into the escape-steering path would be building on a hypothesis, not a result, and this paper’s own standard (§1) is not to do that.

7. The measurement §6 asked for, run

§6 named the exact test this section runs: is real semantic drift closer to the coherent (persistent-bias) regime a P_c pulse can refocus, or the incoherent (memoryless) regime §4 already showed it is inert on. No production session log exists to replay (WP-25 documents precisely what this platform does and does not persist), so this uses the closest honest available proxy: real prose, run through WP-22’s own deterministic lexicon TERA projector (`odu_coverage_map.text_to_tera` — the identical function a live session calls, not a stand-in), paragraph by paragraph in each document’s own original order. Seven sessions, one per markdown source in this paper series (`docs/papers/sources/*.md`, 14 to 77 paragraphs each), 273 paragraphs total — real technical prose written for its own reasons, not authored to make a point about drift.

Method. Per session: compute the TERA sequence $T_1, \dots, T_n$, then the increment sequence $d_i = T_{i+1} - T_i$. The statistic under test is the lag-1 autocorrelation of consecutive increments — per

axis (Pearson correlation of d_i against d_{i+1}) and vector-wise (cosine similarity of consecutive d_i). A positive, significant value means a step tends to continue the same direction as the one before it — the coherent regime. Significance via a real permutation test (5,000 shuffles of each session’s own step order, preserving the exact step-size distribution while destroying temporal adjacency), not a parametric assumption — the same rigor standard as §3’s synthetic model, applied to data this paper did not get to construct.

Result: neither of §3’s two regimes. Every axis and the vector-wise cosine measure show large, overwhelmingly significant **negative** lag-1 autocorrelation (real ≈ -0.3 to -0.5 against a null of $\approx 0 \pm 0.05$, two-sided permutation $p < 10^{-4}$ pooled across all seven sessions on every measure). A step tends to *reverse* direction from the step before it — real technical prose alternates registers paragraph to paragraph (a dense claim followed by a plain-language gloss, a result followed by its caveat) far more than it drifts persistently in one direction. This is real, unambiguous, non-random structure — not the null result a cautious reader might expect from “the first version of this test just didn’t find anything.” It is also, honestly, **not the regime this paper’s mechanism was built for**: not persistent bias (§3’s coherent regime, which P_c cancels), and not noise (§4’s incoherent regime, which P_c is inert on). A mean-reverting process already carries its own restoring tendency; whether injecting a P_c pulse into one would help, do nothing, or actively interfere by forcing early a correction the process was already going to make on its own is a question about a *third* regime this paper’s model does not cover and this measurement does not attempt to settle by itself.

What this does and does not change. §6’s hypothesis — that WP-22’s visit-history pull resembles the coherent regime — is not confirmed by this measurement, on this proxy. That does not mean P_c has no home in this architecture; it means the search for one should not start from the assumption tested here, and a literal production session log (once one exists to replay) could show materially different structure than static prose paragraph order — a real generation loop’s escape-steering redirects are an active intervention this proxy cannot include. The honest scope of this section: real text, the real projector, a real significance test, a genuine and unexpected finding — not a closed question about live sessions, which remains open.

8. Reproducibility

`packages/libaamt/experiments/echo_refocus.py`, no arguments, seed fixed in-file (20260708). Runtime: under two seconds on a laptop CPU (numpy, vectorized across all 50,000 trials per cell). Every number in §4 is printed directly by the script, including the pass/fail check against the stated prediction.

§7’s measurement: `maaiam-alchemist/apps/desktop/scripts/measure_ocm_drift_coherence.py`, no arguments, seed fixed in-file (20260711), depends only on `odu_coverage_map.py` + `voxel.py` (numpy, no model, no network). Runtime under a second. Every number in §7 is printed directly by the script.

9. Why the negative autocorrelation: a null-model artifact, not mean-reverting content

§7 reported a real, overwhelmingly significant finding and stopped there, honestly flagging it as unexplained. This section investigates the *why* — and the answer changes how §7 should be read: the negative autocorrelation is real, but the claim that it reflects real prose “alternating register” does not survive a corrected test. It is very likely, primarily, a statistical artifact of how §7’s significance test was constructed, not a property of the seven documents’ actual content or the order their authors wrote them in.

1. The mechanism. First-differencing *any* sequence is a textbook time-series identity, not a claim about text: for a stationary sequence X_t with autocovariance $\gamma(h)$, the increment $d_t =$

$X_{t+1} - X_t$ has $\text{Cov}(d_t, d_{t+1}) = 2\gamma(1) - \gamma(0) - \gamma(2)$ and $\text{Var}(d_t) = 2\gamma(0) - 2\gamma(1)$. For an i.i.d. X ($\gamma(1) = \gamma(2) = 0$), this reduces to $\text{Corr}(d_t, d_{t+1}) = -\gamma(0)/2\gamma(0) = -\mathbf{0.5}$ exactly, regardless of what X represents. Differencing *manufactures* negative lag-1 autocorrelation out of sequences with no temporal structure whatsoever. Verified directly: differencing pure Gaussian noise (no text, no `odu_coverage_map`, no semantics) at each real session’s exact size gives per-session lag-1 autocorrelations of -0.36 to -0.67 ($n = 2000$ reps at $n = 60$: mean -0.490 ± 0.093 , converging on the analytic -0.5) — the same range as §7’s real result (-0.28 to -0.55).

2. Null-model controls (the request this section answers directly). Three controls were run through §7’s *own* permutation test, unmodified, for direct comparability: **(A)** pure i.i.d. $\text{uniform}[0, 1]^4$ noise, bypassing `text_to_tera` entirely; **(B)** the same 273 real paragraphs with their content untouched but their *order* randomly shuffled, through the real `text_to_tera`; **(C)** “word salad” — each session’s own vocabulary shuffled at the word level and re-chunked to the real paragraphs’ own length distribution, through the real `text_to_tera`. All three reproduce §7’s result almost exactly:

control	T	E	R	A	cosine
§7 real order (reference)	−0.456	−0.276	−0.435	−0.549	−0.362
A: pure i.i.d. noise	−0.496	−0.391	−0.448	−0.469	−0.430
B: real text, shuffled order	−0.443	−0.299	−0.495	−0.522	−0.321
C: word salad	−0.453	−0.321	−0.370	−0.503	−0.366

— every control, at $p < 10^{-4}$ on every axis, by the identical test §7 used. A test that returns the same “large, significant” verdict for real prose, randomly-ordered real prose, word salad, and pure noise indiscriminately is not discriminating between coherent and incoherent drift; it is detecting something the differencing operation manufactures on its own. This directly confirms the concern in this task’s second line of investigation: the effect is a property of the mapping-plus-analysis pipeline, not of the corpus’s semantic content.

3. Axis decomposition. WP-22’s lexicon is tuned for mythic/archetypal prose (“wounded healer,” “hollow king”), not this series’ own technical register, so most paragraphs score zero lexicon hits on most axes and land on the identical sigmoid baseline $\sigma(-1.2) \approx 0.2315$: **E** sits at that exact baseline for 95.8% of the corpus’s 283 paragraphs, **A** for 85.9%, **R** for 47.7%, **T** for 35.7%. A series that is overwhelmingly one repeated constant with rare spikes is close to the delta-function limit where the -0.5 differencing identity is realized most exactly — consistent with, but not required for, the effect (Control A shows it is not required: uniform noise with no baseline clustering at all shows the same magnitude). The negative autocorrelation is present on every axis, not concentrated on one.

4. Lag decomposition. The pure-differencing identity makes a sharp, falsifiable prediction distinct from “prose alternates register every paragraph”: an MA(1) process, negative at lag 1 and $\sim$ zero at lag 2 and beyond — not continued alternation (which would show positive lag-2, negative lag-3, ...). Measured on the real sessions: pooled mean lag-1 -0.429 (mean across axes), lag-2 -0.042 , lag-3 $+0.071$ — a sharp collapse to near-zero, inconsistent sign, immediately after lag 1. This is the MA(1) signature, not a period-2 alternation pattern, and rules out the “real prose alternates register paragraph to paragraph” reading §7 offered as the mechanism.

5. The corrected test — the actual verdict. §7’s permutation null shuffles the *already-differenced* increments, which destroys exactly the structure differencing mechanically creates in *any* order — so it cannot tell a real, meaningful order from a random one apart; every one of Controls A–C confirms this. The methodologically appropriate null instead shuffles the *raw* paragraph sequence and re-differences each shuffle (5,000 draws, same N as §7). Under this corrected null, the real document order’s cosine autocorrelation (-0.362 pooled) is **statistically indistinguishable from random paragraph order** (corrected-null mean -0.400 ± 0.036 , two-sided $p = 0.283$). Individually, six of seven sessions are non-significant ($p = 0.09$ to 0.82); the seventh (wp-23-aamt-substrate.md, $p = 0.010$) does not survive a Bonferroni correction for seven tests ($\alpha = 0.05/7 \approx 0.0071$), and one nominal exception out of seven is within chance expectation.

Conclusion. §7’s “large, overwhelmingly significant negative lag-1 autocorrelation” is real as a *number* — it replicates cleanly on pure noise — but is not evidence that these seven documents’ true paragraph order carries the “mean-reverting/oscillating” content structure §7 proposed. The finding traces to a null-model construction bug: comparing differenced real data against a shuffled-*increments* null cannot detect order-dependence, because differencing induces the same -0.5 -scale signature under *every* ordering. Corrected, the true paragraph order is not distinguishable from a random shuffling of the same paragraphs. This does not resurrect either of §3’s two regimes — the corrected test finds no significant structure of *any* sign, which is closer to §4’s incoherent/memoryless case than to persistent bias, but on a text proxy this section continues to regard as weaker evidence than a real session log (§7’s caveat still applies in full). What §7 called “a genuine and unexpected finding” is, on inspection, a genuine and unexpected finding about the significance test, not about semantic drift — the same standard this paper set for itself in §1: check the precondition before building on it, including the precondition of the test itself.

10. Reproducibility (§9)

maiam-alchemist/apps/desktop/scripts/investigate_negative_autocorr.py, no arguments, seed fixed in-file (20260711), depends only on odu_coverage_map.py + measure_ocm_drift_coherence.py (numpy, no model, no network). Runtime approximately 20 seconds on a laptop CPU (dominated by the 5,000-draw corrected permutation test’s per-draw re-differencing). Deterministic: two independent runs produce byte-identical output. Every number in §9 is printed directly by the script.

Reference implementation: packages/libaamt/experiments/echo_refocus.py, maiam-alchemist/apps/desktop/scripts/investigate_negative_autocorr.py. Series: WP-23 (mirror pairing §10, GF(16) redundancy §11, CRDT convergence §12), WP-22 (Odu coverage map, the projector §7 measures against), EXP-01 (topological trit memory, the series’ first deterministic-seeded-simulation paper).